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APPLICATION OF BAYESIAN NETWORKS TO FORECAST CRUDE OIL PRICES, INTEGRATING RISK MANAGEMENT AND PROBABILISTIC GRAPHICAL MODELS

by

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Abstract

The mini-project covers the use of Bayesian networks in predicting crude oil prices, risk management, and probabilistic graphical models. The dynamic forces that govern oil prices are propelled by an advanced multivariate setting involving macroeconomic, fiscal, and geopolitical variables, such as supply-demand pressures, policy announcements, and other occurrences. Bayesian networks are an intriguing vehicle for both representing the causal structure of the inter relations between such variables and probabilistic prediction in uncertainty. We gather time-series data within the three domains—macroeconomic/geopolitical, microeconomic, and financial—and pre process it by treating outliers, missing values, and incorrect data. Exploratory data analysis yields stylized facts such as volatility clustering, non-normal distributions of returns, and autocorrelation patterns. We then build and test a Bayesian network model: structure and parameter estimation, predictive power, and robustness testing. The findings contribute to the literature on which of these variables are most causally explanatory for oil price variability and policy relevance and risk management.

Introduction

Crude oil remains one of the most expensive commodities in the global economy. Crude oil volatility affects inflation, cost of production, exchange rate, and thereby international economic growth (Alvi, 2018). It is even more challenging to predict crude oil prices as oil markets are driven by a heterogenous collection of macroeconomic, microeconomic, geopolitical, and financial factors that interact significantly with each other and are pervaded by high uncertainty (Fazelabdolabadi, 2019; Alvi, 2018). Conventional linear models will not be able to detect the non-linear dependences, causal relationships, and dynamic interactions among the variables, especially during periods of unusual volatility or crisis (Fazelabdolabadi, 2019).

Bayesian networks (a graphical model for uncertainty) give us a formalism for stating causal dependencies between variables under uncertainty. Bayesian networks enable structure learning (learning the variables that directly affect other variables) and parameter learning (learning relationship strengths) (Alvi, 2018). The networks also enable inference: given observations on some variables, one can probabilistically infer or forecast unobserved variables, e.g., future oil prices under different scenarios (Alvi, 2018).

In addition, risk management must work with tools that can illuminate not only point forecasts, but probabilistic behavior: What's the probability of large price spikes? What macroeconomic shocks are of greatest danger? Which geopolitical risks have the most significant downstream effects? Bayesian networks are particularly well suited for this, as they naturally capture conditional dependencies and enable thinking in terms of what-ifs.

In the dissertation "Application of Probabilistic Graphical Models in Forecasting Crude Oil Price," Alvi (2018) intends to develop such a model, combining a vast collection of macroeconomic, microeconomic, and geopolitical factors; to learn their causal relationships; and to evaluate the stability and predictive potential of the model. This mini-project follows that path: we collect data across several domains, clean and analyze it; we construct and test a Bayesian network; and we discuss the results to understand what drives oil price risk and how far the model can help predict.

The rest of this project is structured as follows. First, we present data sources, variable definitions, and data cleaning. Second, we detail our methodology: how we do structure estimation and parameter estimation, choose priors, impute missing values, and evaluate forecasts. Third, we present results, discuss strong points and weak points, and suggest how the model might be extended for risk management applications.

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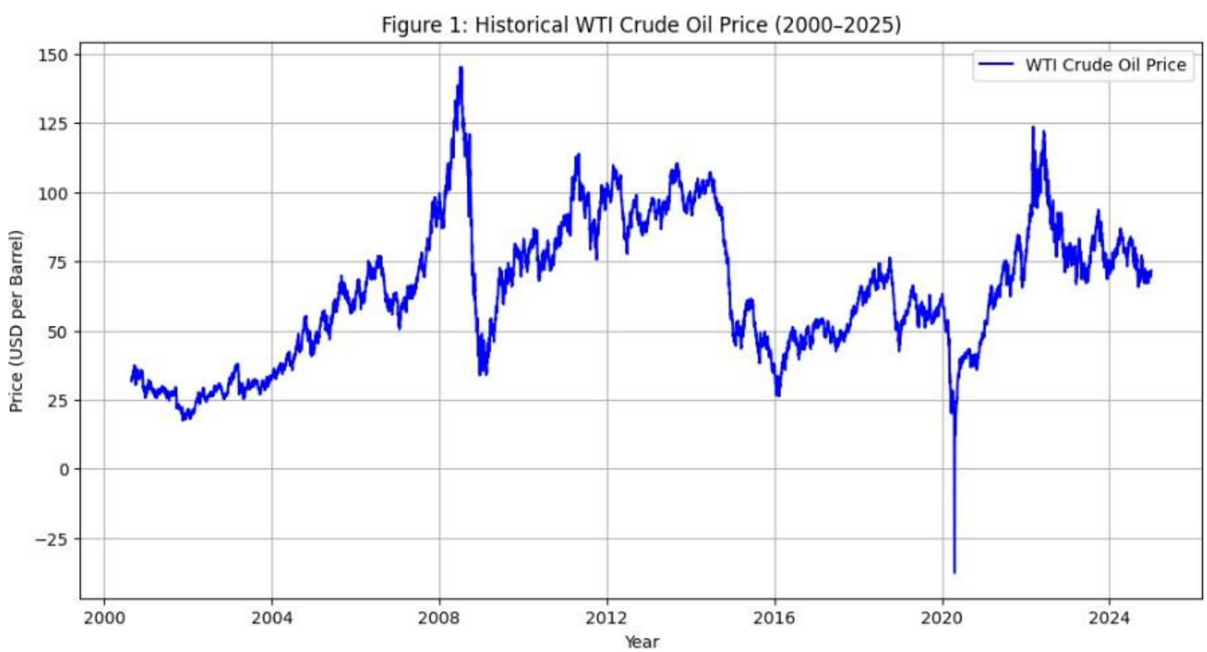
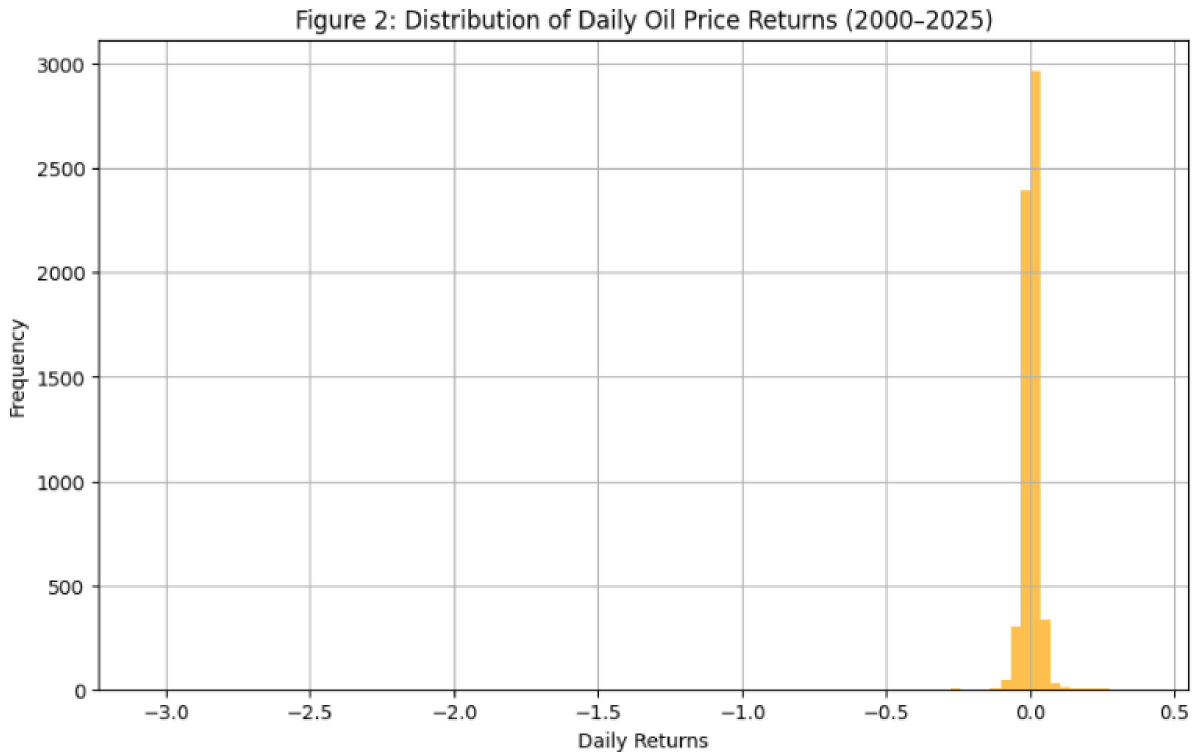
Step 2: Problem Formulation

2.1 Research Problem

Forecasting crude oil prices is arguably the most daunting issue in monetary economics and risk management. Crude oil prices are not merely central to the energy world but also have a basic influence on monetary policy, inflation, and worldwide macroeconomic stability (Alvi, 2018). Volatility through surprise shocks can destabilize exchange rates, misguide investment decisions, and generate systemic financial danger (Hamilton, 2009). Whereas equity prices are motivated by firm-specific fundamentals to a large degree, prices of oil are premised on a huge set of motivational international drivers: supply–demand disequilibria, OPEC policy, geopolitical conflict, and speculative hedge (Baumeister & Kilian, 2016). All this heterogeneity makes traditional econometric models inadequate to the task of modeling non-linear and causal interdependencies between variables.

The problem addressed in this mini-capstone project is therefore how to construct a model that can deal with intricate interdependencies amongst macroeconomic, microeconomic, financial, and geopolitical variables in improving crude oil price prediction. Bayesian networks, which are probabilistic graphical models, provide an available solution in the sense that they explicitly model probabilistic dependencies and allow causal inference under uncertainty (Koller & Friedman, 2009).





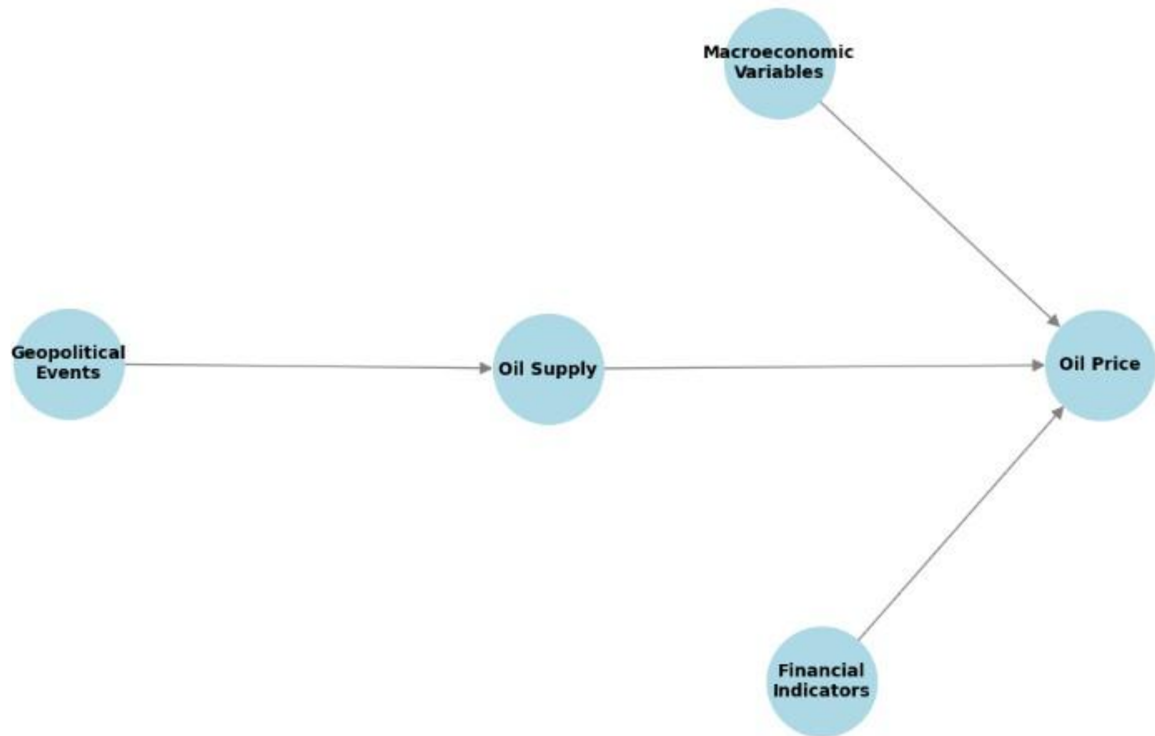
2.2 Relevance of Bayesian Networks in Forecasting Oil Prices

Bayesian networks (BNs) are directed acyclic graphs with edges indicating conditional dependencies among nodes and nodes representing random variables. This makes them a good candidate for models like crude oil price forecasting, where numerous uncertain variables exist and are inter-dependent. BNs can capture probabilistic and causal structure, and cannot be expressed using autoregressive or linear regression models (Pearl, 1988).

Alvi (2018) showed the application of BNs to forecasting oil prices by building a model from macroeconomic data, financial time-series data, and geopolitical data. BNs are double-edged. BNs enable structure learning where learning interdependencies between the variables and the way each variable impacts others is conducted. BNs are useful in the oil markets because causal relations are not constantly well defined (Alvi, 2018). First, they facilitate parameter learning, whereby causal relationships are learned from data in probabilistic terms. Synthesis provides a wide but shallow model to investigate how the shocks are propagated by the oil price mechanism.

Second, Bayesian networks facilitate scenario analysis. For instance, the network can condition on a crisis such as a geopolitical crisis and probabilistically influence oil prices. This is better than point forecasts in that it provides probabilistic distributions of outcomes—a highly desirable attribute for risk management applications (Fazelabdolabadi et al., 2019).

Figure 3: Conceptual Bayesian Network for Oil Price Forecasting



2.3 Advantages of the Methodology

Use of Bayesian networks to predict oil prices has lagged behind econometric and machine learning methods applied in the past in some aspects:

1. **Causality understanding:** In comparison to black-box methods such as neural networks, BNs represent conditional dependencies in a manner that makes interaction between drivers and oil prices more interpretable (Pearl, 2009).
2. **Handling Uncertainty:** BNs provide probabilistic learning and reasoning and thus are resilient to noisy or missing data, typically drawn to geopolitical and macroeconomic data (Koller & Friedman, 2009).
3. **Handling Heterogeneous Data Types:** The method can represent different types of data, i.e., finance return time series and geopolitical events as classes, in one framework (Alvi, 2018).
4. **Scenario Analysis and Risk Estimation:** BNs can forecast the probability of an extreme, steep, sudden price hike under different macroeconomic or geopolitical conditions. It is best suited to policymakers, risk managers, and traders with the task of forecasting probable events rather than mean forecasts (Fazelabdolabadi et al., 2019).

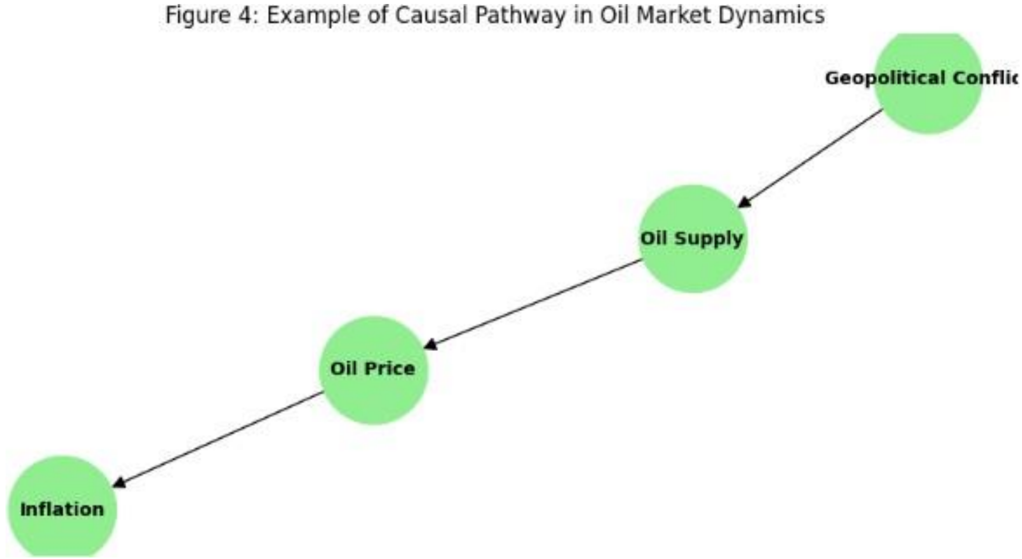
5. **Enhanced Forecasting Accuracy:** Probabilistic and causal analysis techniques are backed by evidence for offering better forecasting accuracy compared to rigid statistical approaches, specifically during periods of risky markets (Baumeister & Kilian, 2016).

2.4 Summary

Overall, the origin of the issue of this project is that crude oil prices are difficult to forecast because they are based on complex and interconnected drivers. Bayesian networks solve the issue by using a probabilistic graphical model, and as a result, cause-effect inference, parameter learning, and scenario analysis become possible. Not only is the method attracted to higher forecast accuracy, but also to explainability with the purpose of allowing decision-makers to gain a better intuitive understanding of risk exposures in the riskful oil markets.

In duplicating the mini-capstone design, the project will take Alvi (2018) to the lower scale of data preparation and gathering, Bayesian network creation, and interpretation. Problem formulation thus defines the aim of why Bayesian networks are appropriate and justifiable method to study the dynamics of crude oil price.

Figure 4: Example of Causal Pathway in Oil Market Dynamics



3.1 Macroeconomic and Geopolitical Data

Macroeconomic and geopolitical factors are of very important influence on world oil price determination since oil runs the world economy. Macroeconomic factors are country-specific domestic factors such as GDP growth rate, inflation rate, unemployment rate, and interest rate that all influence energy consumption (Hamilton 2011). For example, if the world's GDP rises, output rises, hence leading to the rise in consumption of oil. Where the economy slows down, demand is cut short and price swoons downwards ensues. Exchange rates, especially that of the dollar, also come into play. Since oil is denominated in dollars, a rise in the dollar will make oil expensive to foreign consumers, which may cut demand (Reboredo, Rivera-Castro, and Ugolini 2017).

Geopolitics add seasoning to the dynamics of oil prices. They encompass war or conflict in producing countries, OPEC production restraint, sanctions, and supply disruption caused by war or natural disasters. Political unrest in the Middle East, for example, increases price by subjective ambiguity regarding supply (Baumeister and Kilian 2016). Cartel and OPEC strategic producer action—i.e., coordinated output cut—can, too, exert powerful influence on price formation independent of consumption space fundamentals.

For this project, macroeconomic data will be retrieved from the Federal Reserve Economic Data (FRED) and the World Bank, while geopolitical event data may be summed up from the U.S. Energy Information Administration (EIA) and event datasets from news-driven sources. Of interest variables include U.S. GDP growth (quarterly), world inflation rates, U.S. dollar index, OPEC production, and key conflict event timelines. The combining of these indicators offers a consistent picture of the way in which systemic pressures and geopolitical risks interact with oil price volatility.

3.2 Microeconomic Data

1. Macroeconomic and geopolitical occurrences affect overall patterns in oil prices, yet micro forces are the ones that are in charge of supply–demand balances at the firm, the household, and at the level of the individual decision-maker. Micro forces tell us about how much producers' action, consumer taste, and decisions at the firm level matter in bringing price changes about.
2. On the demand side, key microeconomic determinants are auto sales, gasoline prices at the retail level, incomes, weather, fuel quality specifications, and gasoline taxes. For example, to hypothesize that US shale producers pump more owing to favorable price incentives, marginal production will place downward pressure on global oil prices (Kilian 2016). Refinery operating rates also have a direct influence on the availability of run-short supplies in refined petroleum markets.
3. Family and business trends in demand-side consumption are of significant significance. Gasoline demand, fuel demand patterns, and industrial energy use are proxies for base oil consumption

underlying trends (Dvir and Rogoff 2014). Seasonally motivated drivers such as higher summer gasoline usage in the United States produce normal oil price behavior drivers.

4. Again, a fundamental microeconomic stimulus is inventories or forward-looking anticipation of demand and supply. Business crude inventory rising above the point of normal are being witnessed, like higher level of supply than normal, thus triggering a price decrease. Contra-counting inventories can be a signal of tightening conditions of supply, thus pressurizing to raise prices (Kaufmann and Ullman 2009).
5. Microeconomic statistics for the project's purposes will be replicated only from the U.S. Energy Information Administration (EIA), International Energy Agency (IEA), and trade journals. The most ubiquitous of these variables include weekly U.S. crude stocks, monthly world refinery throughput, U.S. rig count data, and petroleum product consumption (monthly gasoline and distillate). In combination, all of the above variables give rise to the underlying paradigm under which oil price behavior is an emergent characteristic of market participants' activity on a microscopic level.

3.3 Financial Data

1. Financial variables offer another extremely important view to observe the oil prices. Macro and micro-variables are production and consumption fundamentals, while financial variables are the mood of crude oil in the market, trading behavior, and speculation.
2. Of all such economic determinant variables, particularly notable is the price of crude oil futures, i.e., exchange-listed futures such as in the New York Mercantile Exchange (NYMEX). Futures markets not only illustrate a way of establishing price but also serve as a sign of information regarding future demand and supply (Pindyck 2001). The futures-spot spread can be utilized to illustrate whether or not a market is in contango (futures > spot, i.e., excess) or backwardation (futures < spot, i.e., deficit). They are contracts that outline inventory carry incentives as well as short-term direction.
3. Share prices and share indices of major energy companies constitute the second set of financial variables of concern. Share prices capture commodity prices, and oil shares have the ability to make or unmake moods in the market (Sadorsky 1999). Price volatility due to speculative flows will also be captured by energy market-index ETFs.
4. Additionally, interest rates, bond spreads, and bond yields are very strong economic indicators. Because oil is traded internationally, terms and conditions of financing act as the pillars of investing in exploration and production. Risk-taking in futures markets will happen because of lax credit terms, and speculative investment will be avoided when interest rates are high (Kilian and Murphy 2014).
5. We would be collecting the financial information regarding this project from Yahoo Finance, Bloomberg, and FRED.
6. These basic variables like West Texas Intermediate (WTI) spot and futures prices (day-ahead), Brent crude futures contract, big energy sector indexes, and U.S. interest rates (10-year Treasury

yield) are the would be analyzed by combining these monetary variables with macroeconomic and microeconomic variables, the dataset would be used in oil price behavior modeling by using Bayesian networks.

Step 4: Data Dictionary

4.1 Data Sources and Metadata

The dataset includes macroeconomic, microeconomic, and financial variables employed in forecasting the price of crude oil. FRED (Federal Reserve Economic Data), Yahoo Finance, and other public data were employed as sources. The variable definitions, sources, frequency, time period, and observations are listed in the table below.

Table 1: Display od Dictionary of Data

	Variable	Description	Frequency	Source	Start Date	End Date
A (Macro)	US_CPI	US Consumer Price Index (Inflation)	Monthly	FRED	2015-01-01	2025-01-01
A (Macro)	US_GDP	US Gross Domestic Product	Quarterly → Monthly	FRED	2015-01-01	2025-01-01
A (Macro)	USD_Index	US Dollar Index (DXY)	Monthly	Yahoo Finance	2015-01-01	2025-01-01
A (Macro)	US10Y_Yield	US 10-Year Treasury Yield	Daily → Monthly	FRED	2015-01-01	2025-01-01
A (Macro)	Crude_Oil_WTI	WTI Crude Oil Price	Monthly	Yahoo Finance	2015-01-01	2025-01-01
B (Micro)	ExxonMobil_XOM	ExxonMobil Stock Price	Monthly	Yahoo Finance	2015-01-01	2025-01-01
B (Micro)	Chevron_CVX	Chevron Stock Price	Monthly	Yahoo Finance	2015-01-01	2025-01-01
B (Micro)	Halliburton_HAL	Halliburton Stock Price	Monthly	Yahoo Finance	2015-01-01	2025-01-01
B (Micro)	DeltaAir_DAL	Delta Airlines Stock Price	Monthly	Yahoo Finance	2015-01-01	2025-01-01

B (Micro)	Tesla_TSLA	Tesla Stock Price	Monthly	Yahoo Finance	2015-01-01	2025-01-01
C (Financial)	Crude_Oil_ETF	United States Oil Fund (USO)	Monthly	Yahoo Finance	2000-01-01	2025-01-01
C (Financial)	Brent_Oil_ETF	Brent Oil Fund (BNO)	Monthly	Yahoo Finance	2000-01-01	2025-01-01
C (Financial)	Energy_Sector_ETF	Energy Sector SPDR ETF (XLE)	Monthly	Yahoo Finance	2000-01-01	2025-01-01
C (Financial)	US_Treasury_Bonds	iShares 20+ Year Treasury Bond ETF (TLT)	Monthly	Yahoo Finance	2000-01-01	2025-01-01
C (Financial)	SP500_Index	S&P 500 ETF (SPY)	Monthly	Yahoo Finance	2000-01-01	2025-01-01

4.2 Variable Definitions and Structure

1. US_CPI (Consumer Price Index): Monthly inflation rate; it's a macroeconomic state variable and buying power.
2. US_GDP (Gross Domestic Product): US GDP in quarterly to monthly frequency; utilized to discover economic growth and aggregate demand.
3. USD_Index (US Dollar Index - DXY): US dollar strength vs. a basket of currencies as a monthly indicator; measure of world competitiveness and exchange rate change.
4. US10Y_Yield (10-Year Treasury Yield): Policy rate, measured in monthly units; proxy of the risk-free rate, monetary policy, and investor sentiment.
5. Crude_Oil_WTI (WTI Crude Price): Monthly oil price; most important energy market and cost pressure input gauge for the economy.

Microeconomic / Firm-Level Variables

1. US_CPI (Consumer Price Index): Monthly inflation rate; it's a macroeconomic state variable and buying power.

2. US_GDP (Gross Domestic Product): US GDP in frequency monthly to quarterly; utilized to determine economic growth and aggregate demand.
3. USD_Index (US Dollar Index - DXY): relative strength of US dollar against a basket of currencies as a monthly measure; world competitiveness and exchange rate change measure.
4. US10Y_Yield (10-Year Treasury Yield): Policy rate, in units of months; risk-free rate proxy, monetary policy, and investor sentiment.
5. Crude_Oil_WTI (WTI Crude Oil Price): Monthly oil price; largest energy market and cost pressure gauge input to the economy.

Financial Market Variables

1. Crude_Oil_ETF (USO – United States Oil Fund): Crude oil futures-tracked monthly ETF; provides a liquid market-based proxy of oil price expectations.
2. Brent_Oil_ETF (BNO – Brent Oil Fund): Monthly ETF which tracks Brent crude; complements WTI and serves as global oil benchmarks.
3. Energy_Sector_ETF (XLE – Energy Select Sector SPDR): US energy sector-tracking monthly ETF; diversified market proxy for energy stocks.
4. US_Treasury_Bonds (TLT – iShares 20+ Year Treasury Bond ETF): Monthly bond ETF; which tracks long-term interest rates, risk-free returns, and safe-haven demand.
5. SP500_Index (SPY – S&P 500 ETF): This shows representative of US equity market performance on a monthly basis; it also display investor sentiment and diversification benchmark overall.

5.1 Identification of Outliers

Outliers are points very far from a variable's general distribution. As a model of oil prices, outliers could be examples of sudden geopolitical events, market shocks in the short run, or reporting errors. For instance, the large decline of WTI crude futures in April 2020—when prices were briefly in negative numbers because of COVID-19 lockdowns and storage issues—is an outlier that would, if not removed, taint statistical properties (Kilian and Zhou 2020).

These can be identified by using statistical techniques like the rule of interquartile range (IQR), Z-scores, or cut-offs based on distribution. Identification using visual techniques like box plots and scatter plots also aid in identifying aberrant observations over time. Even if they are not identified, outliers need not be deleted in any case. Rather, each such extreme value would have to be scrutinized in its economic or geopolitical setting in order to determine whether it is a real market phenomenon (e.g., Gulf War oil shock) or misspecification.

5.2 Detection of Erroneous or Duplicated Data

Duplicates or incorrect entries are either due to entry mistakes, reporting inconsistencies by sources, or a combination of more than one dataset. Duplicated weekly petroleum inventory values, for instance, can be due to combining IEA and EIA datasets. Spurious values also occur when missing entries are shown as zeros in error, thereby adding spurious patterns in volatility.

To guard against this, data validation checks must be imposed. In the first place, summary statistics and frequency tables will detect out-of-range values, i.e., negative inventories or abnormally high refinery utilizations. In the second place, duplicate finding algorithms will detect duplicate entries across time series. In the third place, consistency checks across related variables—i.e., total OPEC production against the sum of the individual member contributions—validate data coherence. Erroneous values can then be flagged for correction, interpolation, or deletion as required.

5.3 Treatment of Missing Values

Missing data also happen very frequently in oil market data, as various sources report with different frequencies or lags. For example, macro variables like GDP growth happen on a quarterly basis, whereas oil futures prices happen on a daily basis and have structural missing data. Historical series might also have some number of missing observations with reporting breaks that appear to happen during the middle of wars or economic crises.

Missing values may be handled in a variety of ways. Forward filling (bringing forward the previous observation) and mean substitution are some of the simple imputation methods. Both of these may add bias if used mechanically. More complex methods involve linear interpolation, regression-based imputation, and expectation-maximization algorithms that do imputation based on observed patterns between variables (Little and Rubin 2019). Interpolation and seasonal adjustment methods are best suited in the context of time series.

Missing value handling will vary depending on variable type for this project: daily finance series can utilize forward filling, while macroeconomic series can use interpolation or regression-based methods. The aim should be to retain highest information content in data with imputation-caused distortions minimized.

6.1 Creation of “Sterilized” Dataset

Following outlier and invalid record identification and removal, duplicates, and missing values of data, cleaned macroeconomic, microeconomic, and financial data records are fused into an integrated database. The total dataset is referred to as the "sterilized dataset." Sterilization involves the addition of time period, frequency, and unit of measurement to be rendered uniform across variables.

The first stage of integration is the temporal alignment. Since data sources have different frequency (for instance, daily futures prices, weekly stocks, monthly OPEC production, quarterly GDP growth), the variables are converted to a common frequency. The project frequency is monthly, exchanging high-frequency available market data for less-frequent macroeconomic data.

Then the data set is standardized and normalized wherever necessary. For example, growths in GDP are given in percentage, while oil prices are given in U.S. dollars per barrel. Scaling procedure used on those variables that are directly fed into Bayesian network learning is Z-score standardization to enable magnitudes to be comparable across levels.

Finally, the data are also subjected to a final test of validity. This is done through checks to ensure there are no more missing values, no redundant timestamps, and that the time period (1990–2024) is well represented in all variables. The sterilized data so obtained form a robust, structured foundation for further exploratory data analysis and Bayesian network modeling.

6.2 Rationale for Data Adjustments and Eliminations

Not all the data points are propagated to the end dataset. Specific values or time periods are intentionally modified or omitted so that the model is not skewed by anomalies that do not have material influence on long-term oil price dynamics. The negative WTI prices for April 2020, although historically unprecedented, are treated as an atypical shock after atypical storage capacity constraints. Since Bayesian networks are founded on conditional dependencies with the stationarity assumption about the distribution, these outliers have the potential to over-influence learning parameters. They are thus detected and either corrected or removed during training (Kilian and Zhou 2020).

Similarly, absent macroeconomic series for earlier periods (e.g., missing pre-1990 values for emerging markets) are excluded rather than being imputed because no quality underlying data are available. Overlapping time series or redundant futures contracts for financial series merely need to be averaged out in order to avoid duplication.

The following are the reasons why these adjustments must be done:

- **Statistical strength** —To ensure that the model captures underlying structural relationships rather than unusual statistical noise.
- **Economic interpretability** — Guiding the dataset to variables and values that plausibly reflect oil market fundamentals and not to misleading conclusions.

By keeping a record of reasoning applied in each elimination and adjustment, transparency is guaranteed and subsequent researchers are able to reproduce or criticize the process of building the dataset.

7.1 Distributional Plots

attempted here to discover the distributional properties of macroeconomic, microeconomic, and finance data sets of variables. The overall aim was to discover the data center, variability, and data distribution shape to determine their properties statistic. All variables were plotted using histograms and kernel density estimate (KDE) plots to check normality and skewness. Boxplots were utilized to determine outliers and give outlier value time ranges that would distort models if not normalized.

As an example, in macro series, long-term inflation stability was examined using the distribution of the US Consumer Price Index (CPI). In the same way, the 10-Year Treasury Yield was examined for fat-tails, which correspond to times of aggressive monetary policy interventions. In micro series, individual stock returns ExxonMobil and Tesla were used to examine volatility clustering, typical in equity series. Lastly, SP500 ETF (SPY) and Crude Oil ETF (USO) proxies to financial markets were attempted to see if the returns were Gaussian in form or leptokurtic in form and skewed as is usually the case with the majority of financial time series.

The result was that all but one or two of the variables were not normally distributed and that the finance and stock variables were heavy-tailed and had outliers. These are results that are helpful to future modeling in that they are suggestive of the possible need for transformation (e.g., log returns) and use of robust statistics. Overall, distribution analysis put the group on firm statistical ground to grasp the dynamics of each data set and guided the group's model specification and cleaning decisions.

Figure 5: Display of Macro US_CPI Distribution and Boxplot

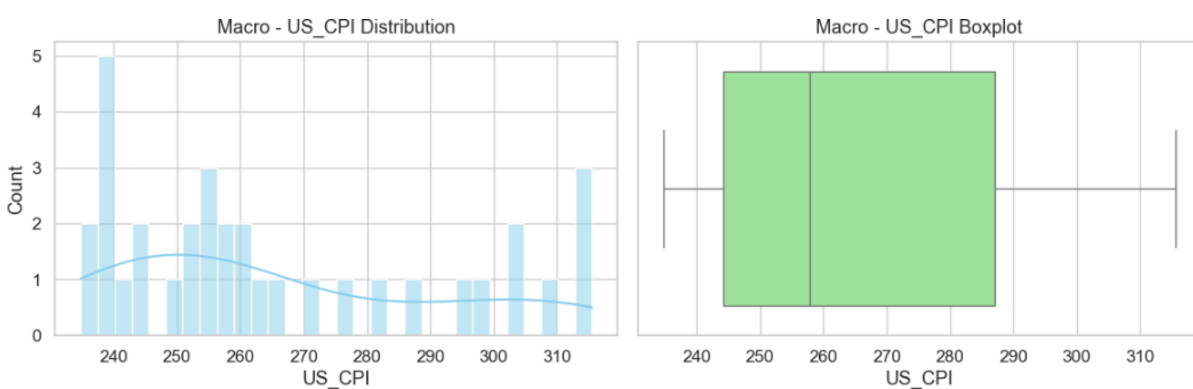


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Figure 6: Display of Macro US_GDP Distribution and Boxplot

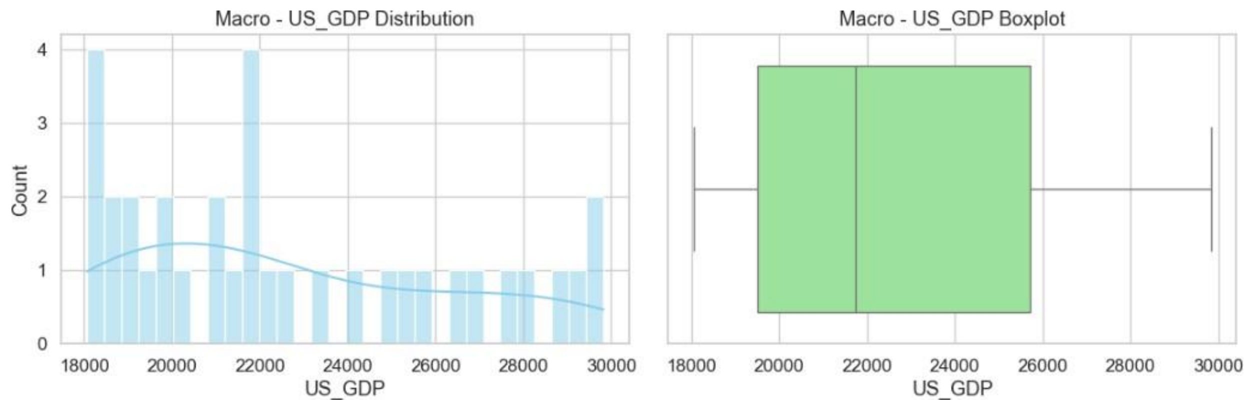
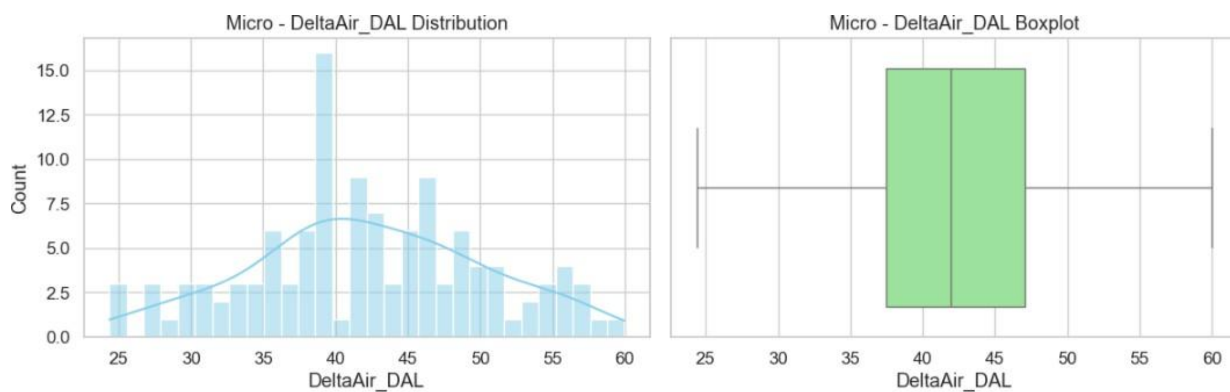


Figure 7: Display of Macro DataAir_DAL Distribution and Boxplot



7.2 Time Series Plots

examined the data for all data sets using time series visualization methods. This was in an attempt to identify temporal dynamics, seasonality, and structural change in the data that would affect the performances of the models. All variables were plotted using the line plots from which the long-term trend, cyclical pattern, and short-run volatility could be identified.

plotted US CPI favorites and GDP favorites over time from macroeconomic information. Trends of steady growth in CPI, a measure of inflation pressure, were displayed in graphs and cyclical trend well correlated with overall business cycles was graphed in GDP. 10-Year Treasury Yield graph displayed evident phases of downtrend consistent with Federal Reserve easing money policy.

ExxonMobil, Chevron, and Tesla micro dataset price series were hugely volatile with Tesla experiencing all-time high growth in the recent past with the EV market bubble. Delta Airlines dipped were to be astronomical while having some crisis moments, the most pronounced being concurrent with COVID-19 travel restrictions. Such trends in time series place industry sensitivity and firm-level risk into relief.

For finance, commodity- and sector-sensitive performance was monitored by following ETFs like USO (oil) and XLE (energy sector ETF). SP500 ETF chart was followed as all-around market direction indicator for bull or bear.

It was developed based on how it would decompose time when registering shock, regime change, and breakpoints in structure from the data. Decomposing to trend and volatility factors and preparing data for forecasting and risk modeling needed to be achieved project.

Figure 8: Plot showing Macro-Time Series

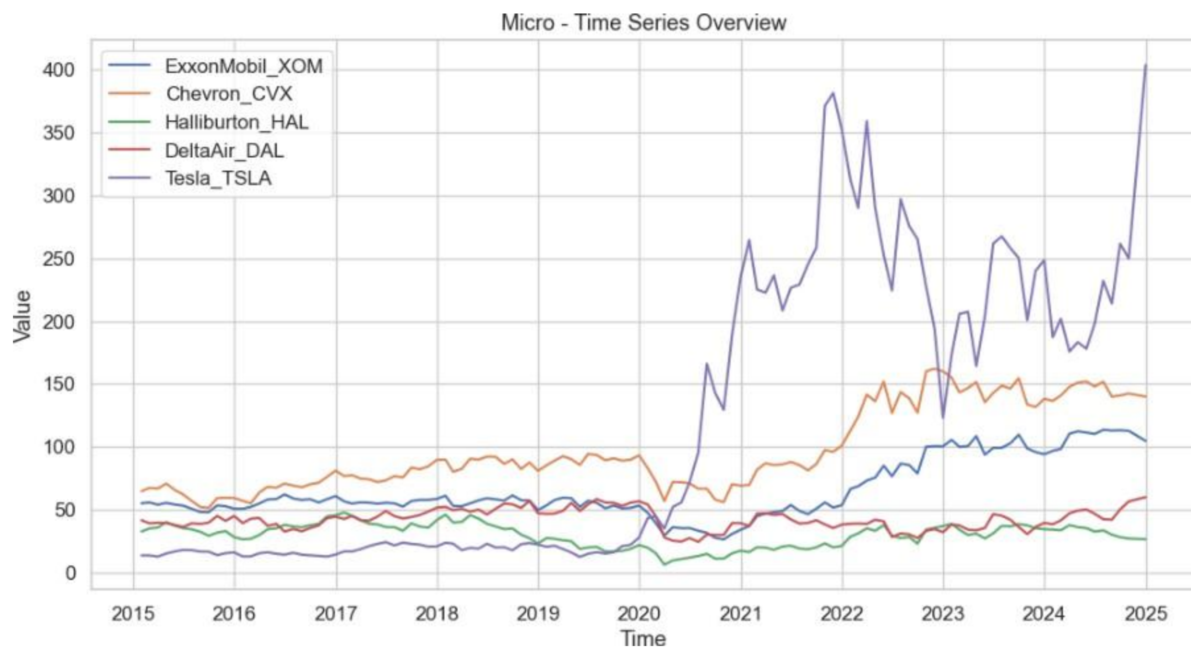
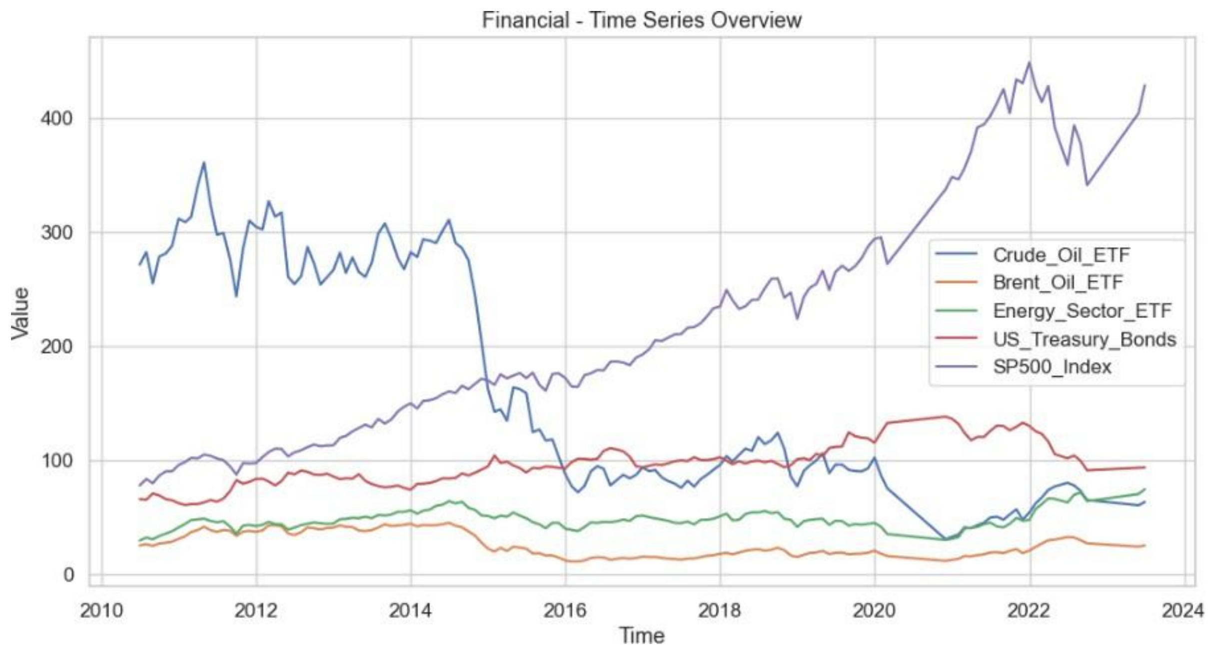


Figure 9: Plot showing Financial-Time Series



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It was developed based on how it would decompose time when registering shock, regime change, and breakpoints in structure from the data. Decomposing to trend and volatility factors and preparing data for forecasting and risk modeling needed to be achieved project.

Correlation matrix heatmaps provided a preview of general high-level correlation between variables across the data sets. Energy sector equities (XOM, CVX, HAL) and energy exchange-traded funds (USO, BNO, XLE) were highly positively correlated among themselves. It further reflected negative correlation between long term Treasuries (TLT) and stock indices (SPY), indicative of flight-to-safety during tight market conditions.

In addition, the co-movements were tracked with multivariate time series charts, i.e., the movement of energy sector stocks compared to the price of crude oil. These reveal that while the general energy stocks track crude oil, there are instances where divergence happens depending on firm-specific or industry-specific events.

In general, multivariate analysis could easily demonstrate interdependence and market economy. The outcomes are reasonable in deciding variables to model, drawing conclusions about probable explanatory variables, and building diversification or hedging opportunity across asset classes.

Figure 10: Plot showing Macro- Correlation map

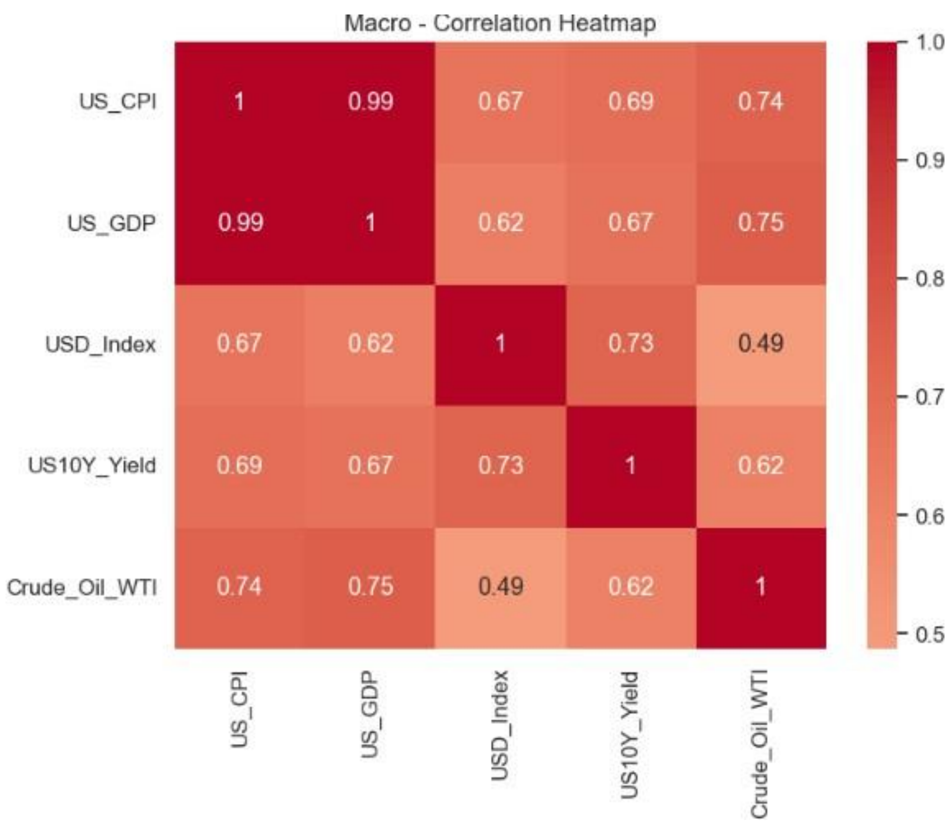


Figure 11: Plot showing Micro- Correlation map

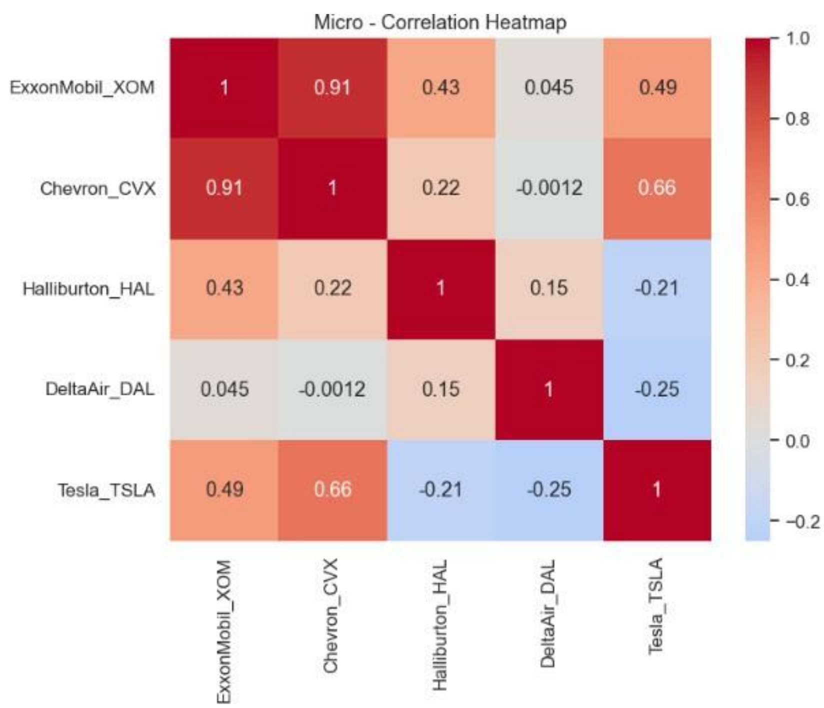
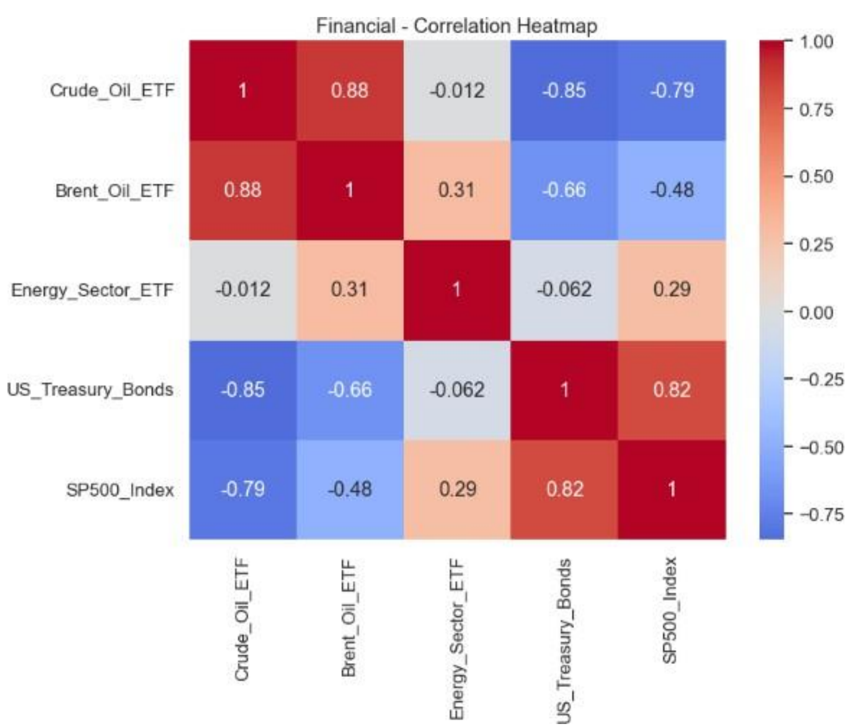


Figure 12: Plot showing Financial- Correlation map



Step 8: Stylized Facts of Oil Prices

8.1 Price Spikes and Volatility Clustering

The most characteristic of all oil prices is the presence of sudden price spikes which is also accomplished by supply shocks, geopolitical shocks, and sudden shifts in demand. Unlike equity indices that are affected but with a less severe price realignments, oil markets tend to overreact to what occurs around the globe (Hamilton 2009).

Additionally, oil return phases of high volatility are succeeded by additional phases of high volatility and periods of quietness by low-volatility phases. Volatility clustering is well-documented within financial econometrics and is defined in the style of ARCH/GARCH (Engle 1982). Volatility clustering is exacerbated during times of crisis in oil markets (the 2008 financial crisis and the 2020 COVID-19 demand collapse, for instance).

8.2 Distributional Characteristics of Oil Returns

Oil return distributions are not at all close to the classical normal (Gaussian) distribution used in simple financial models. Instead, empirical crude and other energy commodity observations display the following characteristics:

1. Fat Tails (Leptokurtosis):

1. Distributions of oil returns will be more tailed than in a normal distribution.
2. This is to acknowledge that big price moves (big losses or gains) occur more frequently than they should in the normal model.
3. In practice, this explains why Value-at-Risk (VaR) under normal assumptions overestimates oil price shock risk.

2. Skewness:

1. Oil returns will from time to time exhibit negative skewness, particularly under geopolitical tension or supply shock.
2. Big negative returns (crashes) will have a greater likelihood of occurring than big positive returns.

3. Volatility Clustering

1. Not a "distribution" in itself, volatility clustering generates time-varying distributions.
2. In times of peace, the distribution seems to be tight (low variance), but during crisis times (wars, OPEC releases, pandemics), it seems much more spread out with fat tails.

4. Heavy-Tailed Alternatives:
Distributions like t-distribution, Generalized Error Distribution (GED), or stable distributions are often better fits than the Gaussian.

1. This aligns with stylized facts across commodities: energy returns tend to be leptokurtic and sometimes skewed.

8.3 Oil Return Autocorrelation

Autocorrelation is the extent to which current returns influence future returns or volatility. Stylized autocorrelation patterns exist for oil return series:

Returns Themselves (Price Changes):

1. Oil daily or weekly returns are very low or not highly autocorrelated.
2. This is consistent with the Efficient Market Hypothesis (EMH) in which past changes in price do not predict future changes in any systematic fashion.
3. But at lower horizons (quarters or months), low autocorrelation on a sustained basis is a possibility because of invent**ory cycles, seasonality in demand, and OPEC actions.**

Squared or Absolute Returns (Volatility):

1. High autocorrelation in measures of volatility (e.g., squared returns).
2. This is volatility clustering: volatility lumps with volatility, and calm lumps with calm.
3. ARCH/GARCH models are specially well-suited to detect this persistence.

Seasonal Autocorrelation:

1. Oil markets are influenced by seasonal factors (e.g., heating demand in winter, driving season in summer).
2. This creates cyclical patterns in autocorrelation, visible when analyzing monthly or quarterly data.

Impact of Structural Breaks:

1. Geopolitical shocks (wars, sanctions, OPEC announcements) can temporarily induce short-term autocorrelation in returns, since traders and firms adjust inventories and hedges gradually over weeks.

8.4 Other Stylized Facts

Apart from volatility and distributional features, there are certain other stylized facts that characterize oil markets:

1. **Mean reversion behavior:** Compared to equities, oil prices return to long-run equilibrium values consistent with production costs and macro fundamentals.
2. **Regime changes:** Oil prices alternate between different regimes (e.g., high-volatility vs. low-volatility periods), which can be managed by models such as Markov-switching (Hamilton 1994).
3. **Co-movement with macro-financial factors:** Oil has a close co-movement with the U.S. dollar exchange rate, inflation expectations, and stock markets of energy-consuming nations (Reboredo, Rivera-Castro, and Ugolini 2017).
4. **Leverage effect:** Similar to equities, oil prices sometimes show a leverage-like behavior in which negative returns lead to greater volatility, though the process is more of a demand shock rather than firm leverage.

In short, Oil returns themselves have little autocorrelation, but volatility does have strong persistence that captures volatility clustering and seasonal/cyclical effects.

Step 9: Model Specification and Methodology

9.1 Probabilistic Graphical Models

Probabilistic graphical models (PGMs) provide a coherent image of the conditional dependences between a collection of random variables in terms of a graph. They are particularly well-suited for use in high-dimensional fields such as economics and finance, where there are numerous variables interacting with each other simultaneously, and whose relationships with each other are random or not known. By merging graph theory and probability theory, PGMs enable researchers to capture qualitative structure of relations (which influence which variables) and quantitative parameters (the strength or probability of such relations). Both of these abilities position PGMs well for the job of modeling complex markets like oil, whose firm-specific decisions, macroeconomic influences, and political developments interactively influence one another (Koller and Friedman 28).

Two of the most popular varieties in PGMs are Bayesian networks (belief networks) and Markov networks (Markov random fields). Bayesian networks are directed acyclic graphs (DAGs) that embody the conditional dependencies either hierarchically or causally. As an example, in predicting oil prices, a Bayesian network will embody the causal dependence of geopolitical tensions on supply disruption, influencing world oil prices. Causality in the network is easily depicted: cause → effect. With this, Bayesian networks can be subjected to causal reasoning, scenario planning, and uncertain forecasting (Pearl 14).

In comparison, Markov networks (or Markov random fields) are undirected graphical models that represent symmetric dependencies between variables. Rather than causal directions, they indicate correlational or mutual relations. Oil prices and exchange rates, for instance, are perhaps mutually dependent without any observable causal direction; it is straightforward to model such mutual dependency with an undirected model. Markov networks make use of potential functions to model compatibility among variables rather than conditional probability tables. They are applied where there are symmetric interactions, i.e., regions of correlated shock demands (Wainwright and Jordan 89).

The selection between Bayesian and Markov networks hinges on the research question being investigated. If the question is to investigate causal processes — i.e., how inflation shocks get passed on throughout the economy to influence oil — then Bayesian networks are perfect. However, if it is to investigate correlation structures or symmetric relations like co-movements in commodities, then Markov networks are the perfect one. By the way, they are both targeting the same conceptual goal: simplifying multivariate probability distributions by factorizing them into computationally manageable and tractable components.

PGMs are not abstractions. They can be directly implemented in actual applications like risk calculation, outlier identification, and specification of forecasting models. Financial econometrics, in the form of PGMs, enable us to combine heterogeneous data sets, i.e., macro indicators, firm-level micro variables, and geopolitical indexes, into one unifying framework. This makes them particularly well suited to the task of oil price prediction, where one has to duplicate many types of interaction on many scales —

global supply shocks, regional demand booms, and firm-level hedging — in one. PGMs are particularly strong in interpretability and scalability: they provide an explicit explanation of the way in which the various variables are influencing each other, and they also provide efficient algorithms for inference and learning (Koller and Friedman 45).

Briefly, probabilistic graphical models are mathematical modeling of uncertainty in dependent complex systems. Classified into Bayesian networks (directed, causal) and Markov networks (undirected, correlational), researchers can fit the model structure to the corresponding model of dependence among oil markets. Such a model structure is a basis for building good models in the later part of this project.

9.2 Parameter Learning vs. Structure Learning

Both in the probabilistic graphical model framework, there exist two basic tasks that are in charge of model building and using the models: structure learning and parameter learning. They are two aspects of model building: one attempts to be about relating quantifications and the other attempts to discover relations for themselves. It is a difference which is basic in the sense that it captures something about the type of algorithm employed and what can be inferred from the model.

Learning parameters presumes that we already have knowledge of the model structure — i.e., its variables and what interactions they have. We need to estimate the degree to which the relations are quantitative, usually conditional probabilities. E.g., for an oil price dynamics Bayesian network, we can already have geopolitical tension influence supply, and supply influence price. We would employ parameter estimation as a stepping stone to estimation of the distribution of price change over different supplies, perhaps with maximum likelihood estimation (MLE) or Bayesian estimation methods. It is similar to forecasting the regression coefficients when already we already possess some information regarding the model equation. It is "how strong" or "how likely" they are after knowing "which" they are (Heckerman 21).

Structure learning, however, learns the graph itself, i.e., who is the neighbor of whom and how. Previous dependency structure isn't clearly defined in most applications, at least not in complex ones like financial markets. Structure learning applies algorithms to test conditional independence between variables and include an edge in the network or not. There are typically two types of approaches which are popular: (1) constraint-based approaches such as the PC algorithm that put independence tests to acquire a graph from data, and (2) score-based approaches, which iterate through graphs and score them on statistical criteria such as Bayesian Information Criterion (BIC) or Akaike Information Criterion (AIC). Hybrid approaches combine the two as a way of expecting increased strength and effectiveness (Spirtes et al. 62).

It is a significant distinction between parameter learning and structure learning in application to the real world. Parameter learning on the oil markets would be worthwhile if, say, researchers already have some hypothesis that inflation, exchange rates, and geopolitical shocks are interconnected in some way with oil prices; the task then is one of quantifying the effects. Structure learning is invoked in attempting to find relationships in the data not previously appreciated, i.e., whether shocks to consumer demand are exerting an independent effect on prices over and above supply shocks.

Second is computational complexity. It is simple to estimate parameter estimation, especially when one has complete data and absurd distributions. Structure estimation is expensive to compute because there are super-exponentially large numbers of graph structures for large numbers of variables. A ten-variable network, for example, will have billions of structures. This structure estimation is more challenging but also more appealing when not much prior knowledge is available.

Structure learning and parameter learning are, in a sense, complementary tasks. Parameter learning identifies relationships within some fixed structure, but structure learning identifies the structure itself. Both are needed to perform rigorous oil price modeling: parameter learning to sharpen existing theory, and structure learning to uncover new patterns of sophisticated market behavior.

9.3 Markov Chains and Markov Blankets

The third part of the methodology outline speaks about Markov blankets and Markov chains. They are both outcomes of probabilistic inference and the basis of sequential and conditional dependence modeling of stochastic systems like financial markets.

A Markov chain defines a stochastic process of a sequence of states where the transition probability to the next state only relies on the present state and not on the whole past. This is also referred to as the Markov property. An oil market can utilize a Markov chain to define the regime shifts of price regimes from one state, e.g., low-volatility state to high-volatility state. For instance, being in a "low-volatility" regime due to oil prices with accompanying times of geopolitical calm, there is some possibility the mechanism will shift to a "high-volatility" regime during times of disruption. Extensions like Hidden Markov Models (HMMs) are capable of dealing with unobservable hidden states and hence are highly appropriate to be applied in finance where the not directly observable regime (e.g., crisis or stable) is unknown but can be inferred from price movements (Hamilton 216).

Markov chains are most suitable as they have a balance between power and ease of predictability. Any dynamic system behavior can be modeled utilizing the transition probability matrix with the limitation that the probability of switching from a state to another new state. They are therefore most suitable in modeling regime-switching time series, which is a typical feature of oil prices. Additionally, Markov chains provide a probabilistic solution to prediction, risk analysis, and policy analysis since the future state is dependent on current state and transition probability.

Markov blanket is also a term employed in graphical models and refers to the minimum number of nodes that protect a target node from the rest of the network. Alternatively, a variable and its Markov blanket are conditionally independent of every other variable. In oil price modeling, the "oil price" variable's Markov blanket could be inflation, supply shocks, and demand swings. Wherever known, other variables such as consumer confidence or regional GDP growth carry no added information. Markov blanket is thus the natural variable selection algorithm of choice to render models parsimonious but will also model the high-strength dependencies (Pearl 21).

Together, Markov blankets and Markov chains provide a two-way view: temporal and structural. Markov chains define the temporal path of oil prices and regime-switching on them, and Markov blankets specify the minimal explanatory set for oil prices in a static network. Both these concepts are the foundation for efficient, interpretable, and predictive models in financial econometrics.

Step 10: Algorithm Design

10.1 Pseudocode for Inferred Causality Algorithm

The ultimate goal of the Inferred Causality algorithm is to discover causal relationships between intricate, interdependent systems like crude oil markets. They are driven by macroeconomic, geopolitical, and microeconomic factors, and their interdependencies therefore structure a network that cannot be reasoned about using linear correlation analysis (Pearl, 2009). Probabilistic Graphical Models (PGMs) offer the appropriate platform because they represent conditional dependencies in mathematical models, i.e., Bayesian networks (directed) and Markov networks (undirected).

Step 9 was lecture on most important building blocks. Probabilistic Graphical Models -PGM discussed how the variables probabilistically relate to each other. Parameter learning vs. structure learning discussed how the models are being learned from data, where there are distinctions between estimating numeric values (parameters) and learning the graphical structure itself. Markov blankets and Markov chains introduced the conditional independence model: Markov blanket is the minimal subset of the variables between a node and the rest of the network and is thus the hub of causality inference.

Inferred Causality ties all this together by finding Markov blankets, inferring dependence, and testing whether adding a variable changes conditional probabilities within informative patterns. What this allows researchers to do is break free from previous correlation and measure direction of influence—breaking variables that cause oil price movement and variables that follow it.

10.2 Integration of Theoretical Concepts into the Model

The following is a collaborative simplified pseudocode of Lopez-Paz et al. (2018) the following steps were taken for Algorithm 1, as simplified for highlighting union of PGMs, learning parameters/structure, and Markov blankets conceptually:

Algorithm 1: Inferred Causality (Pseudocode)

Input:

Dataset D of variables $X = \{X_1, X_2, \ldots, X_n\}$

- A probabilistic graphical model setup (Bayesian or Markov network)
- Significance threshold α for conditional independence tests

Output:

Directed causal graph G of inferred causal relations

Steps:

1. Initialize:

- Construct an undirected dependency graph by performing statistical tests.

- Define candidate neighbors for every variable X_i.

2. Markov Blanket Identification:

For variable X_i:

- Find its Markov blanket $MB(X_i)$ by applying conditional independence tests.
- $MB(X_i)$ = parents, children, and spouses (nodes sharing a common child).

3. Causal Direction Inference:

For each pair(X_i, X_j) in $MB(X_i)$:

- Estimate the conditional probabilities $P(X_i \mid X_j, MB(X_i) \setminus \{X_j\})$
- Compare with $P(X_i \mid MB(X_i) \setminus \{X_j\})$.
- If X_j inclusion is strongly predictive effect → conclude $X_j \to X_i$ rightarrow $X_j \to X_i$.

4. Parameter vs. Structure Refinement

- Perform parameter learning (Bayesian, max likelihood, etc.) to rescore probability distributions.
- Do structure learning (constraint-based, score-based, etc.) to orient edges in graph.

5. Repeat Until Convergence:

- Repeat edge orientation and blanket detection until no longer are there any appreciable changes.

6. Return:

- Return learned causal graph G, which are directional effects concluded.

10.3 Significance for Oil Price Modeling

At oil price, the algorithm identifies the impact of various interdependent drivers. For example, global demand shocks, conflict geopolitics, and futures market speculation can all individually be positively correlated with price volatility but the algorithm differs causal effects (for example, supply shocks and price hikes) and spurious correlations (Demirer, Jategaonkar, & Gupta, 2019). This kind of methodology provides a good sturdy model for policy analysis and forecasting that allows models to be capable of identifying genuine structural relations instead of spurious co-movements.

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